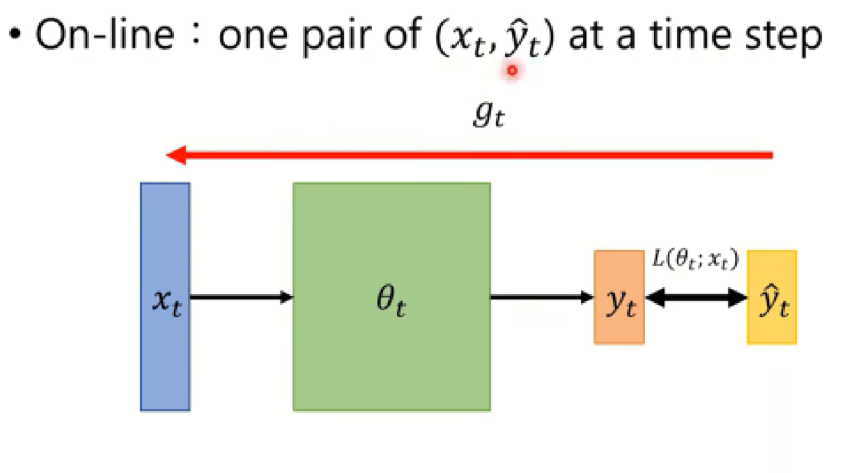
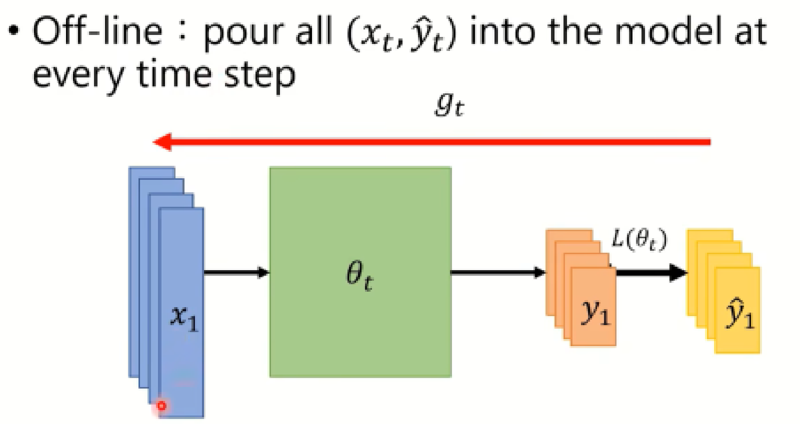
# New Optimazation For Deep Learning

这篇的内容最好搭配相关的论文

some Notations

- $\theta_t$:在第t步的模型参数
- $\nabla L(\theta_t)$ or $g_t$:$\theta_t$的梯度，用来计算$\theta_{t+1}$
- $m_{t+1}$:从第0步到第t步的动量，用来计算$\theta_{t+1}$




- On-line learning:一次处理一次一个x和y
- Off-line learning:一次处理所有的x和y

20年主流模型都在使用Adam和SGDM进行优化

- Adam：fast training, large generalization gap, unstable
- SGDM：stable, little generalization gap, better convergence(?)

An intuitive illustration for generalization gap

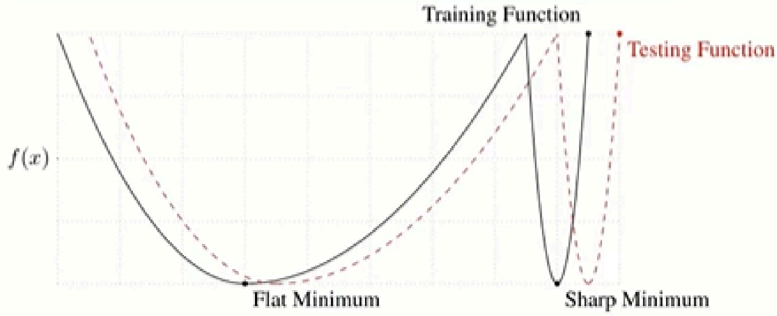



Figure 1: A Conceptual Sketch of Flat and Sharp Minima. The Y-axis indicates value of the loss function and the X-axis the variables (parameters)

Adam能更快的训练，泛化间隙大，但不稳定
SGDM更稳定，泛化间隙小，收敛效果好

## SWATS

**考虑将Adam和SGDM结合**
一开始先用Adam，最后收敛的时候用SGDM

# 考虑优化Adam

## AMSGrad

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t + \varepsilon}} \hat{m}_t$$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_{t-1}, \beta_1 = 0.$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)(g_{t-1})^2, \beta_2 = 0.999$$

*The "memory" of $v_t$ keeps roughly 1000 steps!!*

In the final stage of training, most gradients are small and non-informative, while some mini-batches provide large informative gradient rarely.

| time step | ... | 100000 | 100001 | 100002 | 100003 | ... | 100999 | 101000 |
|---|---|---|---|---|---|---|---|---|
| gradient | | 1 | 1 | 1 | 1 | | 100000 | 1 |
| movement | | $\eta$ | $\eta$ | $\eta$ | $\eta$ | | $10\sqrt{10}\eta$ | $10^{-3.5}\eta$ |

当前面的$g$都很小的时候，在后面即使出现一个很大的$g$，对之后$g$的影响也会很小

- AMSGrad [Reddi, et al., ICLR'18]

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t + \varepsilon}} m_t$$

$$\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$$

考虑记录并使用最大的$v_t$，会出现和Adagrad同样的问题，后期更新过慢

## AdaBound

$$\theta_t = \theta_{t-1} - Clip(\frac{\eta}{\sqrt{\hat{v}_t + \varepsilon}})\widehat{m}_t$$

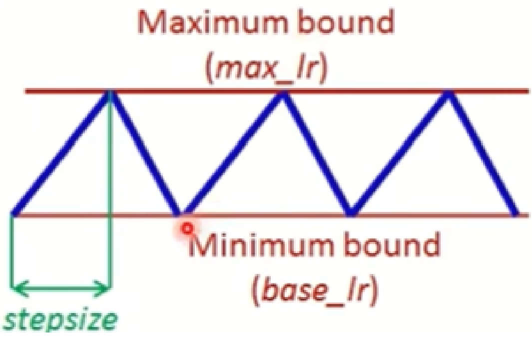
$$Clip(x) = Clip(x, 0.1 - \frac{0.1}{(1-\beta_2)t+1}, 0.1 + \frac{0.1}{(1-\beta_2)t})$$

注：$\beta_2$即Adam中当分母的$\beta_2$
把learning rate做$Clip$，但$Clip$函数是经验公式,lower bound和upper bound已经固定，不算真正的Adaptive learning rate
∴不好用

# 考虑优化SGDM

考虑将learning rate一会调大，一会调小

- Cyclical LR [Smith, WACV'17]
  - learning rate：decide by LR range test
  - stepsize：several epochs
  - avoid local minimum by varying learning rate



- SGDR [Loshchilov, et al., ICLR'17]